ECO 211 — Principles of Microeconomics

Final Exam (Revised)

Name: _____ | **Date:** _____ | **Section:** _____

Instructions: There are 32 questions worth 0.625 points each (20 points total). Mark MC answers by filling the ○ circle, multi-select answers by filling the □ box. For matching questions, write the corresponding letter on each blank line. For ordering questions, write a number from 1 (most) to 4 (least) on each blank line.

Chapter 7 — Production, Costs & Industry Structure

Q1. **[KEPT — Q1 from original]** ORDER — 0.5 pts

Order the following market structures from most competitive (1) to least competitive (4).

_____ Perfect Competition

_____ Oligopoly

_____ Monopoly

_____ Monopolistic Competition

Q2. **[KEPT — Q4 from original]** MULTIPLE CHOICE — 0.5 pts

Given the data in the table below, what will the marginal revenue equal for production at quantity (Q) level 4?

Q	P	TC	TR	MR	MC	Profit
0	$5	$9				
1	$5	$10				
2	$5	$12				
3	$5	$15				
4	$5	$19				
5	$5	$24				
6	$5	$30				
7	$5	$45				

○ $20

○ $1

○ $5

○ $15

Q3. **[KEPT — Q7 from original]** MULTIPLE CHOICE — 0.5 pts

A perfectly competitive firm is currently producing a quantity where Marginal Cost (MC) is $15 and Marginal Revenue (MR) is $20. If Average Total Cost (ATC) is $18, what action should the firm take to maximize profits?

- ○ Decrease output, because ATC is currently higher than MR.
- ○ Decrease output until MR = ATC.
- ○ Increase output, because the additional revenue gained exceeds the additional cost.
- ○ Shut down immediately, as the firm is making a loss.
- ○ Maintain the current output level, as it is earning positive economic profits (P > ATC).

Q4. **[NEW]** MULTI-SELECT — 0.5 pts

Which of the following correctly describe the relationship between a firm's cost curves? Select ALL that apply.

- ☐ MC intersects ATC at ATC's minimum point.
- ☐ MC intersects AVC at AVC's minimum point.
- ☐ When MC is below ATC, ATC is falling.
- ☐ AVC always lies above ATC.
- ☐ When MC is above AVC, AVC is rising.

Q5. **[NEW]** MULTIPLE CHOICE — 0.5 pts

A firm has Total Fixed Costs of $500/week and Average Variable Cost of $8.00 at its current output level. The market price is $6.00 per unit. What should the firm do in the short run, and why?

- ○ Continue operating — fixed costs are sunk and price is still positive.
- ○ Shut down immediately — price is below AVC, so every unit produced deepens losses beyond fixed costs.
- ○ Continue operating — any revenue helps offset fixed costs regardless of variable costs.
- ○ Shut down — price is below ATC, so the firm is making a loss.

Chapter 8 — Perfect Competition

Q6. **[KEPT — Q2 from original]** MULTI-SELECT — 0.5 pts

Select all the conditions for a “perfectly competitive” market structure.

- ☐ No barriers to entry or transaction costs (free entry and exit).
- ☐ No individual seller is able to affect market price on their own.
- ☐ Buyers and sellers have all needed information.
- ☐ There are high quality products made from 1 or 2 providers.
- ☐ Businesses can make the most possible.
- ☐ The government decides who gets what.
- ☐ Many buyers and sellers.

- ☐ All products are identical in that market (interchangeable).

Q7. **[KEPT — Q3 from original]** MULTIPLE CHOICE — 0.5 pts

A small organic farmer operates in a market where the prevailing equilibrium price is $5.47 per bushel for their corn. If the farmer decides to sell their identical corn crop for $6.00 per bushel, what is the most likely outcome?

- ○ The farmer will successfully increase their economic profit margin on every bushel sold.
- ○ The farmer will sell zero bushels because buyers can purchase identical corn for $5.47 from competitors.
- ○ The farmer will capture all buyers currently buying corn at prices below $6.00.
- ○ The farmer's total revenue will increase due to the higher price, despite selling a lower quantity.
- ○ The farmer will still sell their usual quantity because the small change in price does not affect overall market supply or demand.

Q8. **[KEPT — Q5 from original]** MULTI-SELECT — 0.5 pts

An individual competitive firm, in determining its optimal output quantity, must consider which of the following? Select ALL that apply.

- ☐ The price elasticity of demand for the entire industry.
- ☐ The market price for its product, which determines its marginal revenue.
- ☐ The prices of inputs (labor, raw materials) used in production, which determine cost.
- ☐ Whether the market price is sufficient to cover average variable costs in the short run.
- ☐ The quantity where Marginal Revenue equals Marginal Cost (MR = MC).

Q9. **[KEPT — Q6 from original]** MATCHING — 0.5 pts

Match the terms with the correct economic concept. Write the letter of the correct definition next to each term.

Term	Answer	Definition
Small Relative to Market	____	A. Firms have full and transparent knowledge regarding the goods being exchanged.
Perfectly Elastic Demand	____	B. A business can increase or decrease its output without influencing the overall market price.
Perfect Information	____	C. The demand curve faced by the individual firm is a horizontal line.
Identical Products	____	D. The output produced by one firm is highly interchangeable with the

Term	Answer	Definition
		output of any other firm.
Price Taker	____	E. The market pressure forces the firm to accept the equilibrium price.

Q10. **[KEPT — Q8 from original]** MULTIPLE CHOICE — 0.5 pts

In the long run, perfectly competitive firms achieve productive efficiency. This condition is met because:

- ○ The zero economic profit condition means the firms' total revenue exactly equals their variable costs.
- ○ Firms maximize utility rather than profit.
- ○ Firms produce and sell goods at the highest possible marginal cost.
- ○ The competitive process of entry and exit drives the price down to the minimum of the long-run average cost curve.
- ○ P is set equal to MR, maximizing total revenue.

Q11. **[REVISED — based on Q13 from original]** MATCHING — 0.5 pts

Match each economic concept to its significance in monopoly/competition decision-making. Write the letter of the correct significance next to each concept.

Term	Answer	Definition
Price is greater than Marginal Cost (P > MC) at optimal output	____	A. Determines the profit-maximizing quantity of output.
The point on the Demand Curve directly above the MR = MC intersection	____	B. Determines the highest price consumers are willing to pay for the profit-maximizing quantity.
The calculation of Total Revenue minus Total Cost	____	C. Represents the firm's overall highest achievable profit level.
The intersection of Marginal Revenue and Marginal Cost curves (MR = MC)	____	D. A condition demonstrating the lack of allocative efficiency.

Chapter 9 — Monopoly

Q12. **[KEPT — Q9 from original]** MULTIPLE CHOICE — 0.5 pts

A company maintains its dominant market position by controlling access to the only known deposit of a rare mineral input, and also invests heavily in advertising campaigns that make it prohibitively expensive for new firms to gain brand recognition. Which two types of barriers to entry are primarily protecting this firm?

- ○ Trade Secret and Exclusive Licensing.
- ○ Natural Monopoly and Legal Monopoly (Copyright).
- ○ Control of a Physical Resource and Intimidating Potential Competitors (Advertising).
- ○ Natural Monopoly and Predatory Pricing.
- ○ Legal Monopoly (Patent) and Economies of Scale.

Q13. **[KEPT — Q10 from original]** MULTI-SELECT — 0.5 pts

Which of the following legal instruments are specifically designed to promote innovation by granting limited monopoly power over intellectual property? Select ALL that apply.

- □ Trademarks.
- □ Copyrights.
- □ Patents.
- □ Trade Secrets.
- □ Deregulation policies.

Q14. **[KEPT — Q11 from original]** MATCHING — 0.5 pts

Match the correct barrier to entry to each scenario. Write the letter of the matching scenario next to each barrier.

Term	Answer	Definition
Legal Monopoly (Patent)	____	A. Regional water company (duplicative infrastructure costly).
Control of a Physical Resource	____	B. Corporation controls most global mining of a scarce gemstone.
Intimidation (Predatory Pricing)	____	C. Government grants 20 years exclusive right to sell new drug.
Natural Monopoly	____	D. Large retailer sells below cost after competitor enters.

Q15. **[KEPT — Q12 from original]** MULTI-SELECT — 0.5 pts

To calculate and ensure they are maximizing total economic profit, a monopolist must fulfill which of the following conditions or steps? Select ALL that apply.

- □ Determine the quantity of output where Marginal Revenue equals Marginal Cost (MR = MC).

- ☐ Identify the price corresponding to the quantity chosen in step A by referencing the Average Cost curve.
- ☐ Produce where Price (P) equals Marginal Cost (MC) to achieve allocative efficiency.
- ☐ Ensure that price is above the average cost at the profit-maximizing output level.
- ☐ Calculate marginal profit (MR − MC) and confirm it equals zero at the chosen output level.

Chapter 10 — Monopolistic Competition & Oligopoly

Q16. **[KEPT — Q16 from original]** MULTIPLE CHOICE — 0.5 pts

If a monopolistically competitive firm is making positive economic profits in the short run, the process of long-run adjustment ensures that:

- ○ New competitors enter, shifting the existing firm's perceived demand curve to the left until P = AC.
- ○ Existing firms raise their prices to maximize monopoly profits, attracting further entry.
- ○ Barriers to entry increase, allowing the original firm to sustain positive economic profits.
- ○ The firm's marginal cost curve shifts upward due to increased advertising costs.
- ○ The firm increases output until price equals marginal cost (P = MC).

Q17. **[KEPT — Q20 from original]** MULTIPLE CHOICE — 0.5 pts

A key benefit that monopolistic competition offers to society, which is generally absent in perfect competition, is:

- ○ The complete elimination of advertising costs.
- ○ Guaranteed long-run productive efficiency.
- ○ Extremely high barriers to entry protecting firms from competition.
- ○ Guaranteed allocative efficiency (P = MC).
- ○ A desired degree of product variety, innovation, and differentiated shopping experiences.

Q18. **[KEPT — Q21 from original]** MULTIPLE CHOICE — 0.5 pts

When evaluating the tradeoffs of monopolistic competition, the downward-sloping demand curve leads directly to which two efficiency shortcomings?

- ○ Zero economic profit and high barriers to entry.
- ○ High innovation rates and excessive variety.
- ○ Perfect elasticity and low marketing expenditure.
- ○ Productive inefficiency (not min AC) and allocative inefficiency (P > MC).
- ○ Mutual interdependence and the risk of collusion.

Q19. **[KEPT — Q23 from original]** MULTIPLE CHOICE — 0.5 pts

Two rival firms in a duopoly, Alpha and Beta, face a decision: High Output or Low Output. Payoff matrix (profits in $ thousands):

		Beta: Low Output	Beta: High Output
Alpha:	**Low Output**	A = $1,000 / B = $1,000	A = $200 / B = $1,500
	High Output	A = $1,500 / B = $200	A = $400 / B = $400

If both firms pursue their individual self-interest independently, what is the resulting outcome?

- ○ Alpha earns $1,500 and Beta earns $200.
- ○ Alpha chooses Low Output and Beta chooses High Output.
- ○ Both firms achieve the maximum combined profit of $2,000.
- ○ Both firms choose Low Output, resulting in $1,000 profit each.
- ○ Both firms choose High Output, resulting in $400 profit each.

Q20. **[KEPT — Q27 from original]** MATCHING — 0.5 pts

Match each game theory term to its definition. Write the letter of the correct definition next to each term.

Term	Answer	Definition
Game Theory	____	A. The individual strategy that results in the highest payoff regardless of the rival's decision.
Nash Equilibrium	____	B. The result of players maximizing self-interest, often leading to a suboptimal collective result.
Dominant Strategy	____	C. The study of strategic interaction where payoffs depend on other players' actions.
Prisoner's Dilemma	____	D. The characteristic of oligopoly that necessitates the use of game theory analysis.
Mutual Interdependence	____	E. A scenario where cooperation yields larger gains than individual self-interest.

Q21. **[KEPT — Q28 from original]** MULTIPLE CHOICE — 0.5 pts

The crucial concept that makes analyzing an oligopoly market more complex than analyzing a monopoly or perfect competition is:

- ○ The guarantee of zero economic profits in the long run.
- ○ Homogeneous products.
- ○ Low barriers to entry.
- ○ The lack of downward-sloping demand curves.
- ○ Mutual interdependence, where strategic decisions depend on rivals' choices.

Chapter 11 — Antitrust & Regulation

Q22. **[NEW]** MULTIPLE CHOICE — 0.5 pts

An industry has 5 firms with market shares of 40%, 30%, 15%, 10%, and 5%. The approximate Herfindahl-Hirschman Index (HHI) is closest to:

- ○ 500 — well below the 1,000 threshold; unconcentrated market.
- ○ 1,000 — right at the threshold; moderate concern.
- ○ 2,850 — highly concentrated; FTC would likely scrutinize a major merger.
- ○ 10,000 — pure monopoly equivalent.

Q23. **[NEW]** MULTI-SELECT — 0.5 pts

Which of the following are examples of ILLEGAL anticompetitive behavior under U.S. antitrust law? Select ALL that apply.

- □ Three competing airlines secretly agree to charge identical fares on overlapping routes.
- □ A pharmaceutical company earns high profits from a valid patent on a newly developed drug.
- □ Two vitamin manufacturers agree that each will sell only to certain customers, dividing the market.
- □ A large firm prices below average variable cost to drive a new entrant out of business.
- □ A merger between two small firms in a market with more than 50 competitors.

Q24. **[NEW]** MATCHING — 0.5 pts

Match each natural monopoly regulatory approach to its key characteristic. Write the letter of the correct characteristic next to each approach.

Term	Answer	Definition
Unregulated Monopoly	____	A. Theoretically efficient but requires a government subsidy — firm loses money at this price.
Price = Marginal Cost (P = MC)	____	B. Firm sets MR = MC, earns high profit, underproduces — consumers harmed.

Term	Answer	Definition
Price = Average Cost (P = AC)	____	C. Firm covers all costs and earns normal profit — most common practical approach.
Price Cap Regulation	____	D. Regulator sets maximum price for several years; firm profits by cutting costs faster than the cap.

Chapter 12 — Environmental Economics

Q25. **[KEPT — Q29 from original]** MATCHING — 0.5 pts

Match each economic concept related to pollution with its corresponding definition. Write the letter of the correct definition next to each concept.

Term	Answer	Definition
Socially Efficient Quantity	____	A. The sum of a firm's production costs and the additional harm imposed on society.
Social Cost	____	B. The output level where social costs of production equal the benefits perceived by consumers.
Market Failure	____	C. Costs directly incurred by the producer, excluding external impacts.
Private Cost	____	D. A situation where market mechanisms fail to achieve efficient output because not all costs or benefits are accounted for.
Negative Externality	____	E. A consequence of market exchange that negatively affects a third party external to the transaction.

Q26. **[KEPT — Q30 from original]** MULTI-SELECT — 0.5 pts

Which of the following are ways to address a negative externality like pollution? Select ALL that apply.

- ☐ Pollution Taxes.
- ☐ Marketable Permits.
- ☐ Private parties can work out solutions where transaction costs are low and property rights are well-defined.

- ☐ Command and Control Regulations.

Q27. **[KEPT — Q31 from original]** MULTI-SELECT — 0.5 pts

What are shortcomings of Command and Control regulations to protect the environment? Select ALL that apply.

- ☐ They are always cost effective.
- ☐ They offer limited flexibility in where and how pollution will be reduced.
- ☐ They provide no incentive to go beyond the limits they set.
- ☐ They never have politically motivated loopholes.

Q28. **[KEPT — Q33 from original]** MULTIPLE CHOICE — 0.5 pts

A municipality is debating a $10 million project to clean a heavily contaminated lake. Economists estimate the marginal benefit (MB) of the cleanup declines rapidly as water quality improves, while marginal cost (MC) increases dramatically as the last pollutants are removed. Based on efficient resource allocation, when should the municipality stop the cleanup?

- ○ After the first 50% of pollutants are removed.
- ○ When the Marginal Cost of reduction equals the Marginal Benefit of reduction (MC = MB).
- ○ When total estimated benefits first equal total costs incurred.
- ○ Immediately, because $10 million could be spent on life-saving regulations elsewhere.
- ○ Only when marginal cost approaches infinity (100% cleanup achieved).

Chapter 13 — Positive Externalities, Innovation & Public Goods

Q29. **[KEPT — Q35 from original]** MULTI-SELECT — 0.5 pts

Which of the following factors contribute to the private sector's tendency to underinvest in R&D relative to society's needs? Select ALL that apply.

- ☐ The difficulty firms face in fully capturing the total economic value of their successful R&D efforts.
- ☐ The ability of competing firms to adapt or copy underlying ideas without incurring the original R&D costs.
- ☐ The rivalrous and excludable nature of new technologies once successfully implemented.
- ☐ The high fixed costs and inherent uncertainty associated with many R&D projects.
- ☐ The existence of positive externalities, where new innovations provide widespread benefits not captured by the innovating firm.

Q30. **[KEPT — Q38 from original]** MULTIPLE CHOICE — 0.5 pts

A local neighborhood organization relies on voluntary donations to fund enhanced street lighting. If the lighting is nonexcludable and non-rival, the organization will struggle to raise funds due to:

- ○ The free rider problem, where individuals consume the benefit without contributing to the cost.
- ○ The inherent inefficiency that results from providing a club good.
- ○ The high private rate of return that makes governmental intervention unnecessary.
- ○ The tragedy of the commons, where the resource is depleted through overuse.
- ○ The difficulty in defining clear property rights over the service.

Q31. **[NEW]** MULTIPLE CHOICE — 0.625 pts

A profit-maximizing monopolist produces where MR = MC, which results in a price above marginal cost (P > MC). Compared to a perfectly competitive market, this outcome:

- ○ Is productively efficient because the firm maximizes profit
- ○ Creates deadweight loss — output is lower and price is higher than the socially optimal level, reducing total welfare
- ○ Is allocatively efficient because the monopolist serves the market exclusively
- ○ Benefits consumers through economies of scale that lower average costs

Q32. **[NEW]** MULTIPLE CHOICE — 0.625 pts

In the long run, a monopolistically competitive firm produces at a point where price equals average total cost (P = ATC) but NOT at the minimum of ATC. This means the firm:

- ○ Achieves productive efficiency — it produces at the lowest possible cost per unit
- ○ Operates with excess capacity — it could lower average cost by producing more, but demand is insufficient to justify it
- ○ Earns positive economic profit — price above ATC is the long-run equilibrium
- ○ Achieves allocative efficiency — price equals marginal cost at equilibrium

— End of Exam —